Yihao Xu

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Research Profile

- Computational design science researcher working on pre-launch crowdfunding success prediction and multi-view clustering for industrial fault diagnosis.
- First author of a paper at the *International Conference on Information Systems* (2025) and an article in *IEEE Sensors Journal*; the ICIS paper was a Best Complete Paper Nominee / Best-in-Track Paper.
- Experience in neural model design, joint optimization, temporal modeling, and text-based measurement of organizational culture; Python, PyTorch, Java, C, Stata, and SQL.
- Teaching assistant across two fall terms; coordinated six study sessions involving 60 participants who are deaf or hard of hearing in an AR accessibility study.

Education

M.S. in Management Science and Engineering **Expected 2027**
School of Management, University of Science and Technology of China (USTC) *Hefei, China*
Advisor: Prof. Hefu Liu.

B.M. in Information Management and Information Systems **2024**
School of Management, Hefei University of Technology (HFUT) *Hefei, China*
Selected coursework (HFUT grades):
Linear Algebra (96/100, A), Operational Research (95/100, A), Business Data Analytics (91/100, A).

Research Interests

Topics: Digital platforms and crowdfunding · Cultural and creative markets · Organizational culture · Accessible human–AI interaction
Methods: Multimodal representation learning · Multi-view deep clustering · Temporal modeling · Computational text measurement

Published Research

1. Subtly There, Deeply Felt: Modeling Cultural Values in Crowdfunding Success Prediction
Yihao Xu, Wenxiang Zhao, and Xiaoyu Yao.
46th International Conference on Information Systems (ICIS), Sharing Economy, Platforms, and Crowds Track, Nashville, TN, 2025. *ICIS 2025 Best Complete Paper Nominee / Best-in-Track Paper.*

- Creative projects convey cultural meaning through text, images, and metadata. This study models those values and examines their roles in crowdfunding success.
- The Cultural Value-based Prediction Model (CV-PM) maps content features to Throsby's six cultural values. Convolutional networks reduce noise in each value representation, and attention combines the values for prediction.
- On 14,539 Kickstarter projects, AUC reached 94.72% in Art and 95.14% in Design, exceeding the strongest respective baselines by 1.49 and 3.65 percentage points.
- In value ablations, the largest accuracy losses occurred when removing social value in Art (7.17 percentage points) and aesthetic value in Design (5.19).

2. Multisensor Deep Clustering Networks With Joint Learning Strategy for Fault Diagnosis Considering Consensus-Specific Constraint
Yihao Xu, Mingfeng Lu, Gang Wang, and Fang Fang.
IEEE Sensors Journal, 25(11), 19921–19936, 2025. doi:10.1109/JSEN.2025.3560460.

- In unsupervised fault diagnosis, features learned separately from clustering may not suit the clustering task. Multiple sensors also provide consistent and complementary information that the model needs to retain.
- MsDCN uses a sensor-aware attentive encoder to learn consensus representation features and separate encoders to retain sensor-specific features. A consensus-specific constraint guides joint optimization of reconstruction and clustering.
- Across 20 runs per dataset, MsDCN achieved the highest mean clustering accuracy on six gearbox and bearing datasets. Gains over the strongest comparison method were 5.74–11.35 percentage points on the two DIRG bearing datasets. Ablations evaluated the sensor-aware attentive encoder and consensus-specific constraint.

Conference Presentation

Yihao Xu, Wenxiang Zhao, and Xiaoyu Yao. *Subtly There, Deeply Felt: Modeling Cultural Values in Crowdfunding Success Prediction.* 46th International Conference on Information Systems (ICIS), Sharing Economy, Platforms, and Crowds Track, Nashville, TN, Dec. 2025.

Work in Progress

3. Predicting Crowdfunding Success from Culturally Diverse Content in Evolving Environments

Yihao Xu, Wenxiang Zhao, and Hefu Liu. *Planned submission to Information Systems Research in 2026.*

- Introduces a cultural value perspective on pre-launch crowdfunding success prediction, investigating how diverse cultural attributes expressed through campaign text and images jointly inform funding prospects and how their predictive relevance changes as crowdfunding environments evolve.
- Develops SSC-TAP, combining nonnegative least-squares projection onto a cultural semantic cone with hierarchical attention to integrate complementary multimodal information. A temporal model with shared low-rank factors forecasts changes in the crowdfunding environment and each cultural dimension's predictive relevance, adapting predictions to the campaign's content and launch time.

4. From Anchor to Evidence: A Theory-Driven Model for Newly Launched Crowdfunding Success Prediction

Wenxiang Zhao, **Yihao Xu**, and Xiaoyu Yao. *Working Paper.*

- Uses anchoring-and-adjustment theory to distinguish summary cues on listing pages from information on project detail pages when predicting crowdfunding success before launch.

5. Cultural Matching and Supply Chain Efficiency and Resilience

Along Liu, Mengru Zhang, **Yihao Xu**, and Hefu Liu. *Research in progress.*

- Examines how cultural matching relates to supply chain efficiency and resilience among Chinese listed firms and their disclosed suppliers and customers. Measures efficiency through inventory turnover and resilience through changes in cash-flow-based productivity, accounting for firm and year differences.
- Measures cultural orientations in annual-report management discussion and analysis using researcher-defined seed sentences. Qwen LLM embeddings support RAG-style retrieval of related terms from the report corpus to refine seed-sentence weights; cultural matching measures the similarity between supply chain partners.

6. Knowing When to Be Firm: Communication Style Shifts in AI-Assisted Food Delivery

Early-stage Research Project.

- Food-delivery customers may request coupons while leaving emotional needs unspoken. The project examines shifts between assertive and nonassertive AI language during coupon assistance and emotional support. It considers customer preference profiles and responses across conversational turns in relation to subsequent purchase decisions.

Research and Teaching Experience

Research Assistant, University of Science and Technology of China *Jun. 2026–Present*

Supported an AR smart-glasses study on communication access and social inclusion for people who are deaf or hard of hearing; coordinated materials, participant arrangements, device setup, and six study sessions involving 60 participants.

Teaching Assistant, University of Science and Technology of China *Fall 2025, 2026*

Information Technology and Organizational Strategy 42 students; 40 hours

Instructor: Prof. Juan Qin; coordinated assignments, group activities, and feedback.

Awards and Scholarships

Awards

- Outstanding Graduate of Anhui Province (top 3% students) 2024
- Outstanding Student Award (top 10% students) 2023

Scholarships

- China National Scholarship 2025
- First-Class Graduate Scholarship, USTC 2025, 2026
- Shishengqing Scholarship, School of Management, HFUT 2024
- Academic Progress Scholarship, School of Management, HFUT 2022
- First-Class Scholarship, HFUT (top 3% students) 2021

Internships

Research Intern *Jun.–Aug. 2026*

USTC Yangtze River Delta Science and Technology Strategic Frontier Research Center

Conducted desk research, case analysis, and comparative studies of firms' early development and commercialization in commercial spaceflight, AI, and semiconductors.

Technical Service Intern, CEO Office, Volkswagen (Anhui) Automotive Co., Ltd. *Apr.–Jul. 2024*

Hefei, China. Built and maintained an automotive after-sales database for recurring service reporting and operational analysis.

Computational Skills and Languages

Computational Methods: Multimodal learning · Multi-view deep clustering · Hierarchical attention · Temporal modeling · Sentence embeddings and semantic retrieval

Programming: Python · PyTorch · Java · C · Stata · SQL

Languages: Chinese (native) · English (IELTS Academic 7.5)